

SECURITY CLEARANCE: MANUAL-ONLY EXPORT

AELV-1M Technical Material Documentation

Target Profile of the Composite Material AELV-1M

Functional requirements:

- Magnetically recoverable
- Tribo- & piezo-active
- Enzyme-carrying & porous
- Anti-fouling & hydrodynamically structured
- Saltwater stable, mechanically robust

Layer 1 - Magnetic Core (MK-35)

Goal: Magnetic recoverability + mechanical backbone

Material components:

- **Magnetite:** Fe₃O₄ nanoparticles
 - Volume fraction: 35 Vol-%
 - Particle size: 50-150 nm
- **Binder:** PVDF (Polyvinylidene fluoride)
 - Volume fraction: 65 Vol-%
 - Function: Matrix, chemical stability, slight piezo-activity

Example formulation (by volume):

- 35% Fe₃O₄
- 65% PVDF

Properties:

- Density: approx. 3-3.5 g/cm³
- Magnetizable, but only weakly agglomerating without external field
- Completely encased by the motor zone → no direct contact with water

Layer 2 - Motor Zone (MZ-PVDF/Nylon)

Goal: Tribo- and piezoelectric activity + mechanical flexibility

Material components:

- **PVDF:**
 - Mass fraction: 60-70%
 - Function: Piezoelectricity, chemical stability

- **Nylon-6.6:**
 - Mass fraction: 30-40%
 - Function: Triboelectric counterpole, tensile strength

Example formulation (by mass):

- 65% PVDF
- 35% Nylon-6.6

Layer thickness (target range):

- Radial: 5-15 μm around the core

Properties:

- High bending flexibility
- Tribo-contrast PVDF $\leftrightarrow$ Nylon
- Good adhesion to the PAN layer (after surface activation)

Layer 3 - PAN Enzyme Matrix (EM-PAN)

Goal: Porous, enzyme-carrying shell with graded porosity

Material components:

- **Polyacrylonitrile (PAN):**
 - Mass fraction polymer: 100% (without enzymes)
- **Enzyme cocktail (immobilized):**
 - PETase: e.g., 60-70% of the enzyme mass
 - MHETase: e.g., 30-40% of the enzyme mass

Porosity:

- Outer: 0.5-2 μm pores (access for 1-100 μm particles)
- Inner: 50-200 nm pores (enzyme stability, high surface area)

Enzyme loading (target):

- approx. 5-15 mg enzyme per g PAN
- Goal: High enzyme density without diffusion blockade

Layer thickness (target range):

- Radial: 10-30 μm

Properties:

- Chemically stable in saltwater
- High specific surface area
- Good functionalizability (nitrile groups $\rightarrow$ coupling chemistry)

Layer 4 - PTFE Nanostructures (PTFE-NS)

Goal: Charge amplification, anti-fouling, boundary layer braking

Material components:

- **PTFE (Polytetrafluoroethylene):**
 - As nanoparticles or nanorods
 - Particle size: 100-500 nm
- **Coupling agent:** e.g., silane-based fluoro-linkers or plasma activation

Coverage:

- Target: 20-40% of the surface selectively covered with PTFE "scales"
- Not a continuous layer, but islands/spikes

Properties:

- Very high triboelectric negativity
- Anti-fouling (low bioadhesion)
- Generates micro-vortices & braking zones in the boundary layer

Coupling Agents & Interfaces

Critical interfaces:

1. **Magnetic Core ↔ Motor Zone (MK-35 ↔ MZ-PVDF/Nylon)**
 - Solution: Silane coupling agents or functionalized PVDF binders
 - Goal: High shear strength, no delamination during magnetic pull
2. **Motor Zone ↔ PAN Matrix (MZ ↔ EM-PAN)**
 - Solution: Plasma activation of the PVDF/Nylon surface
 - If necessary, thin adhesion layer (e.g., functionalized polymers)
3. **PAN ↔ PTFE Nanostructures (EM-PAN ↔ PTFE-NS)**
 - Solution: PTFE pre-activation (plasma) + silane linkers
 - Mechanical anchoring through rough PAN topography

Geometry & Dimensioning of the Fiber

Typical target geometry:

- Total fiber diameter: 40-80 μm
- Core (MK-35): 10-20 μm
- Motor Zone (MZ): +5-15 μm
- PAN Matrix (EM-PAN): +10-30 μm
- PTFE Structures: in the sub- μm range on the surface

Fiber length in the module:

- Single fiber: mm-cm

- Bundle: fixed in cartridges / grid structures

8. Example: Complete Material Composition per 1 m of Fiber

(Simplified, normalized example)

Assumptions:

- Fiber $\varnothing = 60 \mu\text{m}$
- Density (effective): $\sim 2 \text{ g/cm}^3$
- Volume 1 m fiber: approx. 2.8 mm^3
- Mass 1 m fiber: approx. 5.6 mg

Volume fractions (radially averaged):

- Core (MK-35): $\sim 30\%$
- Motor Zone (MZ): $\sim 25\%$
- PAN Matrix (EM-PAN): $\sim 40\%$
- PTFE Structures: $< 5\%$ (surface only)

Material fractions (rough):

Fe₃O₄:

$$m_{\text{Fe}_3\text{O}_4} = 0.3 \cdot 0.35 \cdot 5.6 \text{ mg} \approx 0.6 \text{ mg}$$

PVDF (Core + Motor Zone):

- Core:

$$m_{\text{PVDF}_{\text{Core}}} = 0.3 \cdot 0.65 \cdot 5.6 \text{ mg} \approx 1.1 \text{ mg}$$

- Motor Zone:

$$m_{\text{PVDF}_{\text{MZ}}} = 0.25 \cdot 0.65 \cdot 5.6 \text{ mg} \approx 0.9 \text{ mg}$$

- Total PVDF: $\approx 2.0 \text{ mg}$

Nylon-6.6 (Motor Zone):

$$m_{\text{Nylon}} = 0.25 \cdot 0.35 \cdot 5.6 \text{ mg} \approx 0.5 \text{ mg}$$

PAN (Matrix):

$$m_{\text{PAN}} = 0.4 \cdot 5.6 \text{ mg} \approx 2.2 \text{ mg}$$

PTFE (Nanostructures):

$$m_{\text{PTFE}} < 0.1 \text{ mg}$$

(Surface fraction)

Enzymes:

(e.g., 10 mg/g PAN $\rightarrow$ 2.2 mg PAN)

$$m_{Enzyme} = 0.022 \text{ mg}$$